

MEASURING CORPORATE EARNINGS DOWNSIDE USING MONTE CARLO SIMULATION AND CORPORATE VALUE-AT-RISK

Scenario Analysis & Hedging Strategy Evaluation

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Executive Summary

Objective & Business Question

This project evaluates the downside financial risk faced by a multinational firm with exposure to foreign exchange rates, commodity input prices, and short-term interest rates, using the Corporate Value-at-Risk (C-VaR) framework. The central business question is not whether volatility exists, but how much earnings downside risk the firm is exposed to under adverse conditions, which risk factors contribute most to that downside, and whether targeted hedging strategies can meaningfully reduce that risk.

The analysis is conducted from a firm-specific earnings perspective, focusing on pre-tax earnings for the year ending 12/31/2024, rather than market value or stock price volatility. This makes the results directly relevant for corporate risk managers, CFOs, and strategic decision-makers.

Data Methodology

The analysis uses quarterly historical data from 1999 to 2023 for three core risk factors:

1. EUR/USD exchange rate
2. Aluminum input prices
3. Short-term interest rates

Historical growth rates are transformed and bootstrapped to generate forward-looking scenarios, preserving real-world distributions without imposing restrictive parametric assumptions. Using Monte Carlo simulation (4,000 scenarios), the model forecasts a full distribution of potential pre-tax earnings outcomes rather than a single point estimate.

Risk is quantified using 95% Conditional Value-at-Risk (C-VaR), which measures the expected loss in the worst 5% of outcomes, providing a more informative view of downside exposure than volatility or traditional Value-at-Risk.

Key Analysis

The project progresses through a structured sequence of analyses:

1. Base Case (Unhedged Firm):

Simulated earnings exhibit meaningful downside tail risk, with C-VaR indicating that extreme but plausible scenarios could materially reduce earnings relative to the benchmark (mean simulated earnings).

2. Stress Scenario (Elevated Volatility):

Increasing the absolute magnitude of historical growth rates by 50% widens the earnings distribution and significantly increases C-VaR, demonstrating how sensitive corporate earnings are to macro-level volatility shocks.

3. Hedging Scenarios:

Separate simulations assess the impact of FX hedging and commodity price hedging by locking in spot values as of 12/31/2023. Each strategy materially reshapes the earnings distribution, but with differing effectiveness in reducing downside risk.

Across scenarios, the analysis highlights that not all sources of volatility contribute equally to downside risk, and that targeted hedging can meaningfully improve the firm's risk profile.

Strategic Implications, Insights, and Conclusions

Three high-level insights emerge from the analysis:

1. Downside risk is economically material, even when expected earnings remain positive, reinforcing the need to evaluate tail outcomes rather than averages.
2. Risk factor asymmetry matters — certain exposures disproportionately drive worst-case outcomes, making blanket risk reduction strategies inefficient.
3. Selective hedging can significantly reduce C-VaR, offering a more capital-efficient approach to risk management than broad volatility suppression.

The results demonstrate how C-VaR can serve as a decision-oriented risk metric, aligning financial risk measurement with strategic planning and treasury decisions.

Tools Used

- Microsoft Excel (Monte Carlo simulation, bootstrapping, scenario analysis)
- Corporate Value-at-Risk (C-VaR) framework
- Historical macro-financial data sources
- Data visualization (earnings distribution histograms, scenario comparison tables)

Business Context & Risk Framework

This section establishes the economic context and risk framework underlying the analysis. Rather than focusing on data mechanics, the objective here is to clarify what risks the firm is exposed to, why those risks matter operationally and financially, and how they translate into earnings uncertainty.

Firm Risk Exposure Overview

The firm operates with meaningful exposure to three primary sources of financial risk:

1. Foreign Exchange Risk (EUR/USD)

A portion of the firm's revenue is generated from European operations and denominated in euros. When revenues are translated into U.S. dollars, fluctuations in the EUR/USD exchange rate directly affect reported earnings. A depreciation of the euro reduces translated revenue, while appreciation increases it. This exposure introduces earnings volatility that is largely outside management's operational control and is driven by macroeconomic and monetary factors.

2. Commodity Input Cost Risk (Aluminum Prices)

Aluminum represents a key input cost in the firm's production process. Changes in aluminum prices affect cost of goods sold and operating margins. Unlike FX risk, which primarily impacts the revenue side, aluminum price movements directly influence costs. Rising commodity prices compress margins, while falling prices improve profitability. Because commodity prices are often volatile and influenced by global supply-demand dynamics, they represent a significant source of downside risk to earnings.

3. Interest Rate Risk (Short-Term Rates)

The firm is exposed to short-term interest rate movements through its financing structure. Changes in interest rates affect interest expense, influencing pre-tax earnings. Although this exposure may appear smaller relative to FX or commodity risk, interest rates tend to rise during tightening cycles that may coincide with adverse macroeconomic conditions, making them an important contributor to tail risk.

Why Pre-Tax Earnings Is the Core Risk Metric

Pre-tax earnings is the most decision-relevant metric for corporate risk management because it links exposures to performance measures that directly affect budgeting, guidance, covenant monitoring, and capital allocation. It also consolidates revenue translation effects, cost variability, and financing costs into a single risk lens that is meaningful for senior management.

Why C-VaR Is Used

Volatility treats upside and downside variation similarly and does not isolate extreme loss scenarios. Traditional VaR identifies a percentile threshold but does not describe how severe outcomes can be beyond that cutoff. C-VaR improves on this by measuring the expected shortfall in the worst 5% of cases, making it better suited for assessing earnings stress and tail-risk exposure.

Data & Modelling Assumptions

This section documents the data inputs and modelling assumptions used to translate historical risk factor behaviour into simulated forward earnings outcomes. The focus is on what data is used, how it is transformed, and the key constraints applied to keep the simulation economically consistent.

Data Sources And Time Period (Quarterly: 3/31/1999 To 12/31/2023)

Quarterly historical time series were collected for the three risk factors that drive the firm's earnings uncertainty:

- EUR/USD exchange rate
- Aluminium price per ton
- 3-month interbank rate

The data window is aligned to the required project horizon and supports a sufficiently large historical sample for scenario resampling.

Transformation Into Growth Rates

Each risk factor is converted into quarterly growth rates to model changes rather than levels:

- FX and aluminium are modelled using percentage changes (growth rates) from one quarter to the next
- Interest rate movements are modelled consistently as quarterly changes in the rate series (treated as the earnings-relevant driver for interest expense)

This step ensures the simulation captures historical variability and co-movement patterns in the form most relevant to forecasting.

Bootstrapping Logic (Sampling Historical Growth Rates)

Scenario generation is based on bootstrapping, where each forward scenario is constructed by randomly sampling quarterly growth rates from the historical dataset and compounding them forward to produce simulated paths. This approach is intentionally non-parametric: it relies on realized historical distributions rather than assuming normality or a specific parametric process. The method preserves empirical features such as skewness and tail behaviour that matter for downside risk measurement.

Key modelling assumptions

The simulation converts risk-factor paths into earnings outcomes using the following firm-level assumptions, designed to reflect operational linkages while maintaining internal consistency:

1. Revenue Translation: Euro-denominated sales are translated into USD using the simulated EUR/USD path
2. Cost Structure: Aluminium-related input costs vary with the simulated aluminium price path
3. Interest Expense: financing costs vary with the simulated short-term rate path
4. Positive Expected Earnings Constraint: Inputs are specified so the benchmark (mean simulated pre-tax earnings) remains positive, ensuring an economically meaningful baseline

These assumptions create a transparent bridge from historical market dynamics to forward earnings distributions, enabling consistent C-VaR estimation across the base case, stress scenario, and hedging scenarios.

C-VaR Methodology & Scenario Design

This section describes the analytical structure used to generate earnings scenarios and quantify downside risk using the Corporate Value-at-Risk (C-VaR) framework. The methodology is applied consistently across all scenarios examined in the project to ensure comparability of results and clarity in interpretation.

Scenario Design & Simulation Structure (4,000 Scenarios)

Pre-tax earnings for the year ending 12/31/2024 are estimated using 4,000 Monte Carlo simulation scenarios. Each scenario represents a distinct realization of future economic conditions affecting the firm. Scenario paths are constructed by randomly sampling historical quarterly changes in each risk factor and compounding them forward over the forecast horizon. This approach generates a broad distribution of potential earnings outcomes while remaining grounded in historically observed behaviour and preserving empirical co-movement across risk factors.

Transmission of risk factors into earnings

For each scenario, simulated risk-factor paths feed into earnings through clearly defined operating and financial channels. Movements in the EUR/USD exchange rate affect the U.S. dollar value of euro-denominated revenues. Changes in aluminium prices influence input costs and operating margins. Variations in short-term interest rates affect interest expense through the firm's financing structure. These effects are combined to generate a single pre-tax earnings outcome per scenario, ensuring internal consistency across simulations.

Benchmark construction

The benchmark used for risk measurement is defined as the mean pre-tax earnings across all 4,000 simulated scenarios. This benchmark represents expected earnings under the modeled risk environment and provides a stable reference point against which downside deviations are evaluated. All C-VaR calculations are expressed relative to this benchmark.

Definition and interpretation of 95% C-VaR

The 95% Corporate Value-at-Risk (C-VaR) is calculated as the difference between the benchmark (mean simulated pre-tax earnings) and the average pre-tax earnings in the worst 5% of simulated outcomes. This measure quantifies the expected severity of earnings losses under adverse but plausible conditions, relative to expected performance.

Unlike volatility or percentile-based measures, Corporate C-VaR focuses explicitly on earnings downside from a firm-level perspective, making it particularly well suited for evaluating corporate financial risk and comparing the effectiveness of alternative hedging strategies under stressed conditions.

Earnings Risk Outcomes and Hedging Impact

Scenario 1 (Base Case) : Unhedged Earnings Risk Profile

Scenario Description

This base case evaluates the firm's pre-tax earnings risk assuming no hedging of foreign exchange rates, aluminium input costs, or short-term interest rates. Using 4,000 simulated scenarios, the analysis generates a forward-looking earnings distribution for the year ending 12/31/2024. This scenario establishes the reference risk profile against which stress conditions and hedging strategies are evaluated.

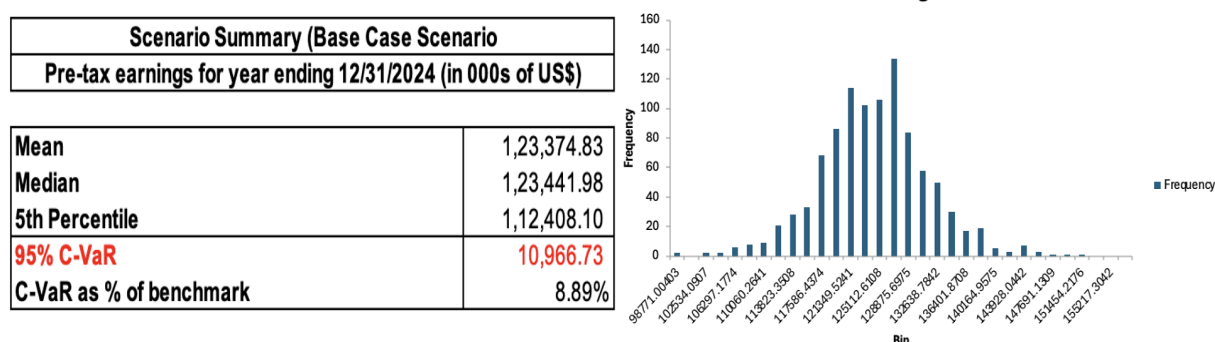


Figure 1 : Simulated Pre-Tax Earnings Distribution and 95% Corporate -VaR — Unhedged Base Case

Earnings Distribution Characteristics

The simulated earnings distribution is centred around a stable benchmark, with most outcomes clustered within a relatively narrow central range. The histogram exhibits a clearly defined left tail, indicating that while extreme downside outcomes are less frequent, they are economically meaningful. The proximity of the mean and median suggests stability in typical earnings outcomes, but dispersion in the lower tail reflects vulnerability to unfavourable joint movements in exchange rates, commodity prices, and interest rates.

The shape of the distribution indicates that downside outcomes are driven not by isolated shocks, but by correlated adverse realizations across multiple risk factors. This interaction effect amplifies earnings stress in extreme scenarios and widens the lower tail relative to what would be implied by independent risk movements.

Downside Risk Assessment

Downside risk is summarized using the 95% Corporate Value-at-Risk, which captures the expected earnings shortfall in the worst 5% of scenarios relative to the benchmark. In the unhedged case, the C-VaR represents a material reduction in earnings, highlighting that tail risk is significant even when expected earnings remain positive.

Risk Interpretation

The base case demonstrates that the firm's unhedged earnings profile is exposed to meaningful tail risk arising from the combined effects of revenue volatility, cost variability, and financing uncertainty. This profile provides a critical baseline for evaluating whether changes in the volatility environment or the introduction of targeted hedging strategies can meaningfully reduce downside exposure without compromising expected performance.

Scenario 2 (Stress Scenario) : Increased Risk Factor Volatility

Scenario Description

This scenario evaluates the firm's earnings risk under a heightened volatility environment by increasing the magnitude of historical quarterly growth rates for all risk factors by 50% in absolute terms. Each observed growth rate is multiplied by 1.5 while preserving its sign. This approach amplifies fluctuations without altering the underlying directional behaviour or historical structure of the risk factors.

The objective is to simulate a stressed macro-financial environment that remains economically grounded and allows assessment of earnings sensitivity to amplified shocks.

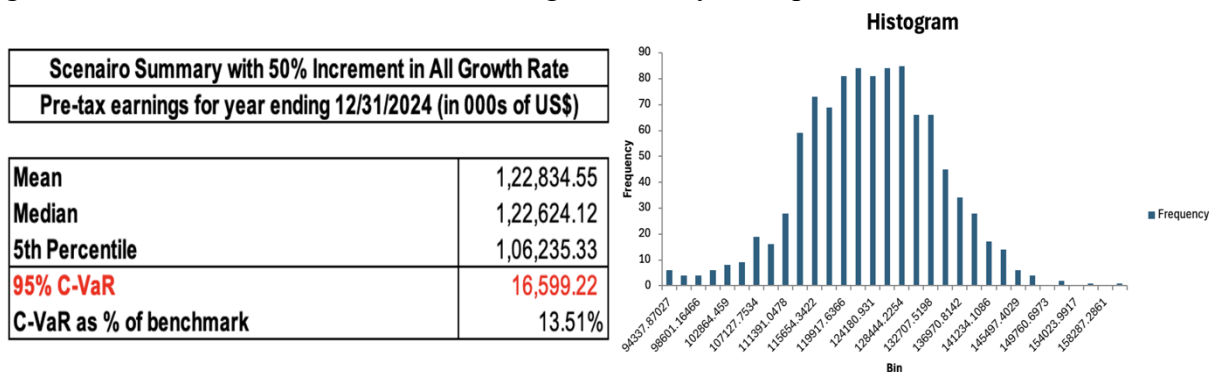


Figure 2 : Simulated Pre-Tax Earnings Distribution and 95% Corporate -VaR — Stress Scenario with Elevated Volatility

Earnings Distribution Characteristics

Relative to the base case, the earnings distribution widens substantially. The histogram shows increased dispersion across outcomes and a more pronounced left tail, indicating that low-earnings scenarios become both more frequent and more severe under stressed conditions. Extreme outcomes extend further away from the benchmark, reflecting the compounding impact of amplified shocks across exchange rates, commodity prices, and interest rates.

The broader spread of the distribution highlights that earnings outcomes become less predictable in high-volatility environments, with a wider range of potential realizations compared to normal conditions.

Downside Risk Assessment

The 95% Corporate Value-at-Risk deteriorates materially compared to the base case, indicating a larger expected earnings shortfall in the worst 5% of scenarios. This increase reflects not only greater volatility in individual risk factors but also stronger interaction effects when multiple adverse shocks occur simultaneously.

Risk Interpretation

The base case demonstrates that the firm's unhedged earnings profile is exposed to meaningful tail risk arising from the combined effects of revenue volatility, cost variability, and financing uncertainty. This profile provides a critical baseline for evaluating whether changes in the volatility environment or the introduction of targeted hedging strategies can meaningfully reduce downside exposure without compromising expected performance.

FX Hedging Scenario: Exchange Rate Locked

Scenario Context

This scenario evaluates the impact of foreign exchange risk management by assuming the EUR/USD exchange rate is locked at its 12/31/2023 level for the entire forecast horizon. Aluminium prices and short-term interest rates remain stochastic, allowing the analysis to isolate the effect of eliminating exchange-rate-driven variability in earnings. The objective is to assess how stabilizing revenue translation influences the firm's downside earnings risk under otherwise unchanged operating and financial conditions.

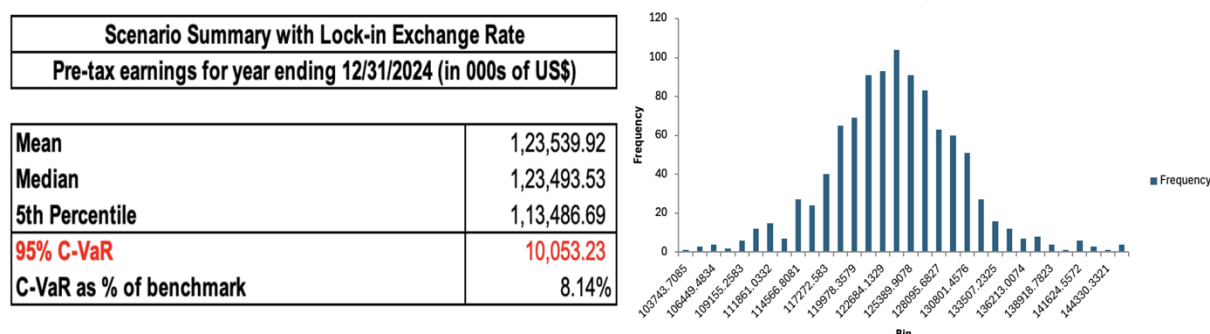


Figure 3 : Simulated Pre-Tax Earnings Distribution and 95% Corporate - VaR — FX Hedging Scenario (EUR/USD Locked)

Earnings Distribution Characteristics

Relative to the unhedged base case, the earnings distribution becomes noticeably tighter. Dispersion around the benchmark declines, and the left tail shortens, indicating fewer extreme low-earnings outcomes. This reflects the removal of uncertainty associated with translating euro-denominated revenues into U.S. dollars. Central outcomes remain largely unchanged, suggesting that FX hedging stabilizes earnings without materially shifting expected performance.

The compression of the distribution also signals reduced interaction effects between exchange rate movements and other risk factors. By removing one major source of volatility, adverse joint realizations across risk factors become less severe, leading to improved predictability of earnings outcomes. However, some dispersion persists due to continued exposure to aluminium prices and interest rate movements.

Downside Risk Assessment (Corporate C-VaR)

The 95% Corporate C-VaR improves materially relative to the base case, confirming that exchange rate movements were a significant contributor to earnings downside risk. The reduction in C-VaR reflects a decrease in the severity of adverse earnings outcomes rather than an increase in expected earnings. This highlights the effectiveness of FX hedging as a downside protection tool.

Risk Interpretation

This scenario indicates that exchange-rate uncertainty was a meaningful driver of left-tail earnings outcomes in the unhedged case. By reducing translation volatility, FX hedging improves earnings predictability and planning reliability. The results also suggest that FX hedging primarily enhances downside protection rather than upside potential. As a result, FX hedging emerges as a targeted and efficient risk management strategy.

Commodity Hedging Scenario: Aluminium Price Locked

Scenario Context

This scenario evaluates the impact of hedging aluminium input costs by fixing the commodity price at its 12/31/2023 level over the forecast horizon. Exchange rates and short-term interest rates remain stochastic, allowing the analysis to isolate the effect of stabilizing a key cost component while maintaining exposure to revenue-side and financing risks.

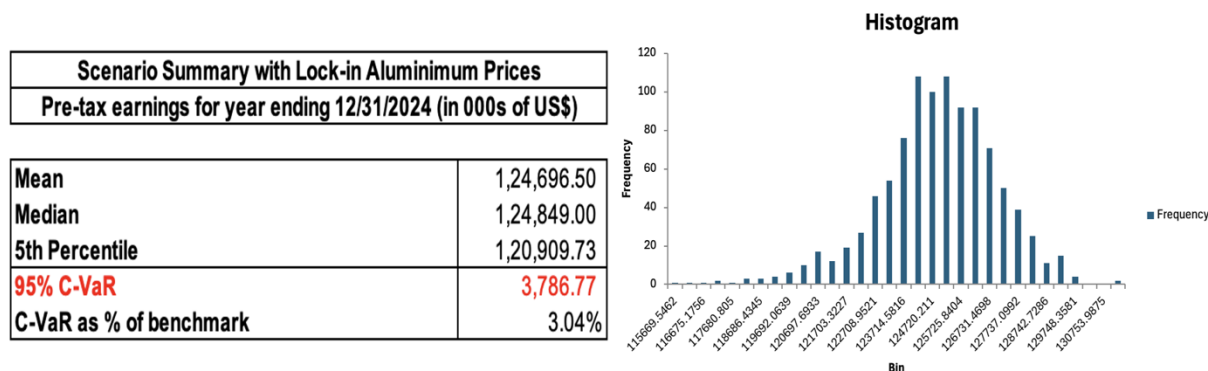


Figure 4 : Simulated Pre-Tax Earnings Distribution and 95% Corporate – VaR — Commodity Hedging Scenario (Aluminium Price Locked)

Earnings Distribution Characteristics

Compared to the unhedged case, the earnings distribution exhibits reduced downside dispersion, particularly in scenarios previously driven by sharp increases in aluminum prices. The left tail narrows, reflecting improved margin stability when input cost volatility is removed. Central outcomes remain broadly consistent with the base case, indicating that cost hedging primarily affects downside risk rather than expected earnings.

However, the distribution remains wider than in the FX-hedged scenario. This suggests that while cost-side volatility is an important driver of earnings risk, revenue translation and interest rate movements continue to influence overall dispersion.

Downside Risk Assessment (Corporate C-VaR)

The 95% Corporate C-VaR improves relative to the unhedged base case, demonstrating that aluminium price volatility contributes meaningfully to downside earnings risk. The magnitude of improvement is more moderate than that observed under FX hedging, indicating that cost shocks are an important but not dominant driver of tail risk.

The remaining C-VaR reflects residual exposure to unhedged revenue and financing risks, which continue to affect earnings outcomes in adverse scenarios.

Risk Interpretation

Commodity hedging enhances earnings resilience by improving margin visibility during periods of rising input costs. However, because downside risk remains from other sources, aluminium hedging alone does not fully stabilize earnings. These results support a portfolio approach to risk management, where cost hedges complement revenue-side and financial risk mitigation strategies rather than acting as standalone solutions.

Robustness Check: Increased Number of Scenarios

Scenario Context

This robustness check increases the number of simulated scenarios (15,000 simulated scenarios) to assess the stability and reliability of the earnings distribution and Corporate C-VaR estimates. The objective is to ensure that the conclusions drawn from earlier scenarios are not sensitive to the number of simulations and are therefore suitable for decision-making.

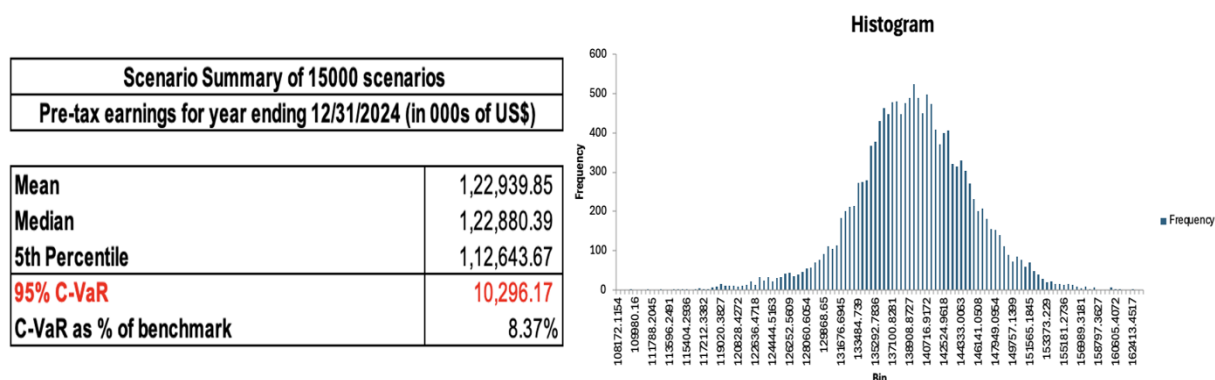


Figure 5 : Simulated Pre-Tax Earnings Distribution and 95% Corporate- VaR — Robustness Check with 15,000 Simulation Size

Earnings Distribution Characteristics

With a larger number of scenarios, the earnings distribution becomes smoother, particularly in the tails, reflecting improved sampling of extreme outcomes. Importantly, the central tendency, overall dispersion, and shape of the distribution remain consistent with those observed in earlier scenarios. No structural shifts are observed in the benchmark or the left-tail behaviour.

The persistence of distributional characteristics indicates that the modeled earnings dynamics are stable and not driven by random resampling effects.

Downside Risk Assessment (Corporate C-VaR)

The 95% Corporate C-VaR remains broadly consistent with prior estimates despite the increase in simulation size. This convergence indicates that downside risk measures stabilize as the number of scenarios increases, reinforcing confidence in the reliability of the estimated C-VaR values.

The relative ordering of scenarios by downside risk remains unchanged, supporting the robustness of comparative conclusions across base, stress, and hedging cases.

Risk Interpretation

The robustness check strengthens confidence in the Corporate C-VaR framework and validates the use of the simulation model for earnings risk assessment. It confirms that observed downside risk characteristics reflect structural exposure rather than simulation noise. From a governance perspective, this supports using the framework for ongoing risk monitoring, scenario updates, and evaluation of alternative hedging strategies.

Conclusions & Risk Management Recommendations

Summary of Findings

This project applies a Corporate Value-at-Risk (C-VaR) framework to evaluate the firm's downside earnings exposure arising from foreign exchange movements, commodity price volatility, and interest rate fluctuations. By modeling a full distribution of pre-tax earnings outcomes rather than relying on a single expected value, the analysis highlights that earnings risk is driven primarily by tail outcomes, not by average performance. Even when expected earnings remain positive across scenarios, adverse but plausible realizations of underlying risk factors can materially impair profitability.

Across scenarios, the base case establishes a meaningful downside earnings exposure, while the increased volatility case demonstrates that corporate earnings risk deteriorates materially when risk factor fluctuations intensify. The hedging scenarios show that selectively reducing exposure to key drivers can compress downside outcomes and improve earnings resilience. Finally, the robustness check confirms that the C-VaR estimates and scenario comparisons remain stable when the number of simulations increases, supporting confidence in the results.

Metric	Base Case	Increased Volatility	FX Rate Lock-In	Commodity Price Lock-In	Robustness (15,000 Scenarios)
95% Corporate C-VaR	10,966.73	16,599.22	10,053.23	3,786.77	10,296.17
C-VaR as % of Benchmark	8.89%	13.51%	8.14%	3.04%	8.37%

Figure 6 : Summary of Corporate Value-at-Risk (C-VaR) Across Scenarios

Hedging Strategy Comparison

The C-VaR summary across scenarios provides clear insight into the relative contribution of different risk factors to corporate earnings risk. Among the scenarios analyzed, the increased volatility case produces the largest deterioration in Corporate C-VaR, underscoring how amplified macro-financial uncertainty disproportionately affects downside outcomes. This highlights that volatility itself is a key driver of earnings risk, independent of changes in expected performance.

Comparing hedging strategies, FX rate lock-in delivers a meaningful reduction in Corporate C-VaR relative to the unhedged base case, indicating that exchange-rate-driven revenue translation is a significant contributor to earnings downside risk. By stabilizing revenue conversion, FX hedging reduces both the severity and dispersion of adverse earnings outcomes while leaving central performance largely unchanged.

Commodity price lock-in produces the largest relative improvement in Corporate C-VaR among the hedging strategies considered. This suggests that aluminium price volatility is a dominant driver of tail risk in the firm's cost structure and that stabilizing input prices significantly improves margin resilience under adverse conditions. The sharper reduction in C-VaR under commodity hedging indicates that cost-side shocks play a particularly important role in shaping extreme earnings outcomes.

The robustness scenario, which increases the number of simulations, produces Corporate C-VaR estimates consistent with the base case. This confirms that the relative ranking of scenarios and hedging strategies is reliable and not sensitive to the number of simulated paths.

Recommendation

Based on the results, a targeted and exposure-driven risk management strategy is recommended rather than a blanket approach to hedging. The analysis indicates that not all sources of risk contribute equally to earnings downside, and that the most effective risk mitigation strategies are those that directly address the dominant drivers of tail risk.

Given the substantial reduction in Corporate C-VaR achieved under commodity price lock-in, priority should be given to managing input cost volatility, particularly in environments where commodity prices are subject to sharp upward movements. FX hedging also provides meaningful downside protection and should be considered where revenue translation risk is material and highly correlated with broader macroeconomic stress.

Importantly, hedging decisions should be framed around reducing downside earnings exposure relative to the firm's benchmark, rather than maximizing expected earnings. The Corporate C-VaR framework offers a practical tool for aligning hedging policy with earnings stability, budgeting confidence, and risk tolerance. Management should use this framework to evaluate hedge ratios, select instruments, and periodically reassess exposures as market conditions evolve.

Limitations and Further Considerations

While the analysis provides a robust assessment of corporate earnings risk, several limitations should be acknowledged. First, the simulation framework relies on historical growth rates and assumes that past behaviour is informative about future dynamics. Structural changes in markets, correlations, or firm operations could alter the risk profile going forward.

Second, the hedging scenarios abstract from implementation costs, liquidity constraints, accounting treatment, and basis risk associated with real-world hedging instruments. In practice, these factors may affect the net benefit of a given strategy and should be incorporated before execution. Additionally, the analysis considers static hedging assumptions; dynamic or layered hedging strategies may further influence outcomes.

Finally, the focus on annual pre-tax earnings does not explicitly account for cash flow timing, liquidity pressures, or covenant constraints that may become binding during stressed conditions. Future extensions could integrate these dimensions, incorporate alternative stress regimes, or model time-varying correlations to enhance decision relevance.

Despite these limitations, the results demonstrate that Corporate C-VaR provides a powerful and decision-oriented framework for evaluating earnings risk, comparing hedging strategies, and supporting informed corporate risk management decisions.